



SiAir: Drone as a Service (DaaS) Last-Mile Food Delivery Logistics in Valais

GROUP 1

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1 Introduction

This report investigates a pivotal logistics question: **can autonomous aerial systems optimize last-mile food delivery while simultaneously cutting environmental emissions?**

Focusing on the alpine terrain of Sierre (Valais), we model a hybrid logistics architecture that pairs bicycle couriers for first-mile collection with heavy-payload drones for distribution to remote mountain communes. The framework evaluates a pilot launch projected for 2030, assuming a matured Swiss regulatory landscape and broad social acceptance of commercial drones. For a seamless reading flow, all underlying assumptions, mathematical models, and technical data are organized in the Appendix.

1.1 Context

Nowadays, food delivery services are primarily operated by cars, or where possible in urban centers, by cargo bicycles or electric scooters. However, when considering remote, high-altitude mountainous areas, the service is exclusively performed by Internal Combustion Engine (ICE) vehicles or completely discontinued due to operational complexity. [1].

Switzerland is specifically vulnerable to these delivery shortfalls; approximately 60% of the national territory is covered by the Alps, a baseline that spikes to over 80% within Canton Valais [2]. In Valais, numerous villages are spread across rugged mountainous terrain where traditional distribution models are limited by extensive travel distances, safety detours, low population density, and impervious alpine road conditions. These topographic constraints make conventional last-mile food delivery economically inefficient and logistically prohibitive. As a result, residents of smaller, remote communes, frequently experience complex access to restaurant delivery platforms, facing delayed or cancelled deliveries, despite being located only a few kilometers away in a straight-line geographic radius.

While laws and regulations are still under development, ongoing commercial drone experiments and pilot operations, such as the recent automated last-mile distribution trials executed by players like Amazon in Italy [3], strongly indicate that this technological disruption generates profound operational advantages. The structural friction separating straight-line geographic distance from road route distance is where a Drone-as-a-Service (DaaS) logistics system becomes relevant.

Drones can fly straight from a centered depot to the customers up in the mountains, cutting out all the road complexities and detours that make normal car deliveries inefficient. Furthermore, it allows restaurants in the valley to easily reach high-altitude villages that were out of range before, making the whole system finally viable. This creates a better logistic system which not only improves the current service, but also broadens the market of food deliveries. On top of that, replacing gas-powered cars with electric drones means cutting out fossil fuels completely. Since the Swiss power grid relies heavily on clean energy like hydropower, it drastically cuts down last-mile carbon emissions without slowing down delivery times. Our assumption matches current research on last-mile logistics, which shows that drones improve local access, lower operational costs, and boost sustainability [4].

1.2 Goal and Scope of the Project

The goal of this project is to assess how a centralized Drone as a Service system could improve last-mile food delivery in the district of Sierre. By combining bike-based order collection with drone (T-Drone M1500 hexacopter [15]) home delivery, the project aims to connect restaurants in Sierre to surrounding communes, expand access to food delivery in less served areas, and design an efficient delivery network adapted to the mountainous context of Valais. The service area includes the communes of Anniviers, Chalais, Chippis, Crans-Montana, Grône, Icogne, Lens, Noble-Contrée, Saint-Léonard, and Sierre.

On the **supply side**, the system is limited to three restaurants located in the same neighborhood in Sierre: McDonald's, Burger King, and Wok Royal.

On the **demand side**, the project estimates food delivery demand at the commune level using population data and food delivery usage assumptions. Demand is then distributed across the week and operating hours, through manipulation using Google Maps data, to represent when orders are likely to [A.5](#). Delivery is also included within the commune of Sierre, as it represents the highest-demand area. Even though Sierre already has better restaurant access than remote villages, drone delivery could still be relevant by reducing short-distance road trips, absorbing peak-hour demand, and offering faster delivery during busy periods. This allows the model to capture both local demand in Sierre and delivery demand from surrounding communes, especially in areas where conventional delivery is limited by distance, terrain, or low population density. The service operates whenever one of the restaurants is open, totaling 100 hours per week, based on the restaurants' opening hours. Orders are first collected by a biker, who performs regular pickup rounds every 15 minutes between the restaurants and one central drone hub; the biker has also the role of swapping batteries and loading drones, this task accounts for an other 5 minutes. From this hub, drones complete home delivery across the district. The fleet will then depart for their delivery mission and the biker will start another 15' round across the 3 restaurants.

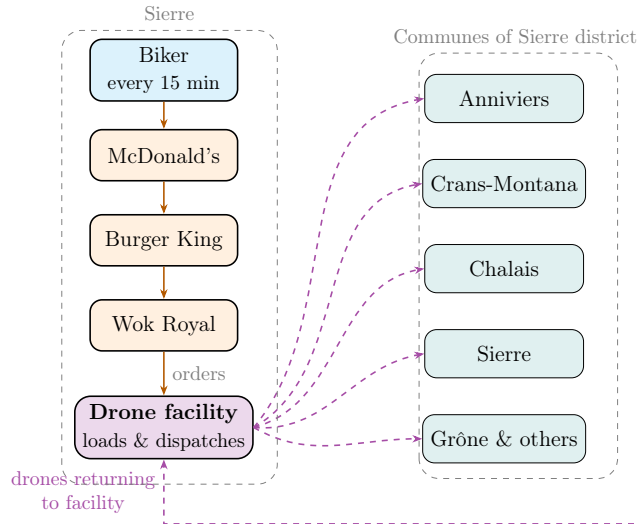


Figure 1: Overview of the hybrid delivery system: the biker visits each restaurant in sequence to collect orders, then brings them to the drone facility, from which drones complete home delivery across the district before returning to base.

1.3 Modeling Assumptions

To keep the optimization model tractable and realistic, the system is bounded by a series of foundational hardware, operational, and financial assumptions. These are structured as follows:

1.3.1 Drone Hardware and Physical Specifications

1. **Fleet homogeneity:** All drones in the system are identical units benchmarked against the T-Drones M1500 hexacopter (15 kg maximum payload capacity, IP55 environmental protection) [15]. No mixed-fleet configurations are considered.
2. **Technical Specifications and Innovation:** All hardware characteristics are extrapolated from the aforementioned commercial drone model to maintain realistic baseline parameters. For specifications not publicly disclosed on the manufacturer website (such as precise battery cell dimensions), values were defined following a standard assumption of technical innovation, ensuring they remain capable of covering the physical distance to the furthest commune considered. The full technical specification matrix is detailed in Table 4 in the Appendix.
3. **Volumetric Cargo Bounds:** The physical volume of the food orders is explicitly neglected in this framework. The system operates on the assumption that any order meeting the weight limits will physically fit inside the cargo bay, bypassing the need to solve a volumetric bin-packing problem.
4. **Payload Unit Proxy:** The average payload per customer delivery is modeled as a fixed unit of $\bar{w} = 1$ kg. This metric uses a standard, fully packaged fast-food menu (e.g., a McDonald’s Big Mac menu inclusive of a main burger, side, large beverage, and bagging insulation) as a realistic weight proxy.
5. **Customer Destination Access:** It is assumed that all end-consumer locations feature an open, unobstructed private or public landing space (such as a garden, driveway, or flat clearance zone) sufficient to allow the drone to complete its automated touchdown and drop-off cycle safely.

1.3.2 Flight Physics and Routing Inputs

1. **Energy Model Linearity:** Per-segment energy consumption is treated as linear in relation to distance, payload weight, and altitude changes (see Appendix A for the full mathematical derivation). Both ascent and descent segments are assumed to draw energy equally, as electric braking systems are modeled to dissipate energy as heat rather than recovering it via regenerative braking. The altitude term therefore utilizes the absolute value $|\Delta h_{ij}|$.
2. **Drone Allocation:** Drones are not all available each 15-minutes round. Each time they are assigned to a delivery route, we compute how much time the travel for the deliveries and coming back to the warehouse take, and we add a 5 minute service time (e.g changing batteries, deliveries loading, quick reparation).

1.3.3 Operational Windows and Lifecycles

1. **Utilization and Service Windows:** The system is analyzed under two distinct schedules. Baseline assumes a full continuous delivery window from 10:00 AM to midnight (extended to 1:00 AM on Fridays and Saturdays), totaling 100 available operating hours per week. Scenario 1 restricts operations strictly to daily lunch and dinner peak hours (12:00 PM–2:00 PM and 6:00 PM–10:00 PM every day), totaling 42 available operating hours per week. Any built-in downtime accounts for automated battery swaps, localized weather holds, and regulatory constraints.
2. **Drone Lifetime and Replacement:** Each drone frame is rated for a certified mechanical lifetime of 4,000 active flight hours [16]. This boundary implies a full vehicle replacement cycle roughly every ~ 1 year under Baseline Scenario and every ~ 2 years under the lower-intensity Scenario 1. Equipment CAPEX is amortized on a per-flight-hour basis accordingly.
3. **Warehouse costs:** The depot expenses include an ongoing fixed cost (monthly rent and utilities) and an initial capital cost (CAPEX) for the ground control equipment and telemetry screens needed for the remote supervision of the drone fleet. The numbers can be found in the Appendix.

1.3.4 Personnel and Economic Parameters

1. **Mission Supervision Ratio:** Following modern Beyond Visual Line of Sight (BVLOS) multi-drone regulatory frameworks, operations do not require a 1:1 pilot ratio. Instead, a single remote technician can supervise a swarm of up to five automated drones simultaneously from a ground station. The control ratio $M/5$ is treated as a continuous variable within the cost function equations [10].
2. **Fixed Biker Courier Support:** The fleet includes a fixed ground element consisting of $N_b = 1$ dedicated bicycle courier across all scenarios. The courier handles deliveries that are technically or regulatory impossible for autonomous aircraft (such as indoor drop-offs, apartment complexes, or deep regulatory no-fly zones). The courier wage is set at a standard local logistics rate of 25 CHF/hour.
3. **Maintenance Charges:** Annual vehicle maintenance is budgeted at 3,000 CHF per drone per year [17]. This yields a consistent operational charge of $3,000/52 \approx 57.7$ CHF per drone per week, covering mandatory scheduled diagnostic inspections, critical part swap-outs (propellers, bearings), and official safety compliance recertifications.
4. **Electricity Tariff:** Power usage costs are priced against the official Swiss Federal Electricity Commission (*ElCom*) industrial mean tariff baseline of 0.27 CHF/kWh (0.27 CHF/kWh) [18]. This energy unit rate is held constant across both configurations to maintain a controlled comparison.

1.3.5 Demand

1. **Static Market Demand:** The model assumes that the current food delivery demand remains static throughout the evaluated operational timeline. We do not account for future market growth, population shifts, seasonal tourist fluctuations in

the Sierre-Anniviers region, or changes in customer ordering habits. The calculated demand acts as a fixed baseline representing current market conditions.

2. **Visual Data Extraction (Pixel Counting):** Because exact quantitative visitor numbers per hour are not publicly disclosed by Google Maps, the relative hourly affluence levels for the target restaurants were extracted manually using a pixel-counting methodology. We measured the exact vertical pixel height of the relative activity bars provided on the visual interface of the platform, assuming a perfectly linear relationship between bar height and customer foot traffic [A.5](#).

2 Mathematical Modeling

2.1 Problem Definition

The core operational problem is modeled as a **Capacitated Vehicle Routing Problem (CVRP)** with energy constraints and time-discretized fleet availability.

Let $\mathcal{V} = \{0, 1, \dots, n\}$ be the set of nodes, where node 0 denotes the drone facility (depot) and nodes $1, \dots, n$ represent customer delivery locations across the communes. Let $A = \{(i, j) \mid i, j \in \mathcal{V}, i \neq j\}$ be the set of directed arcs connecting nodes, and $x_{ij} \in \{0, 1\}$ equal 1 if a drone travels directly from node i to node j , and 0 otherwise.

Every 15 minutes, a new set of orders $\mathcal{P}$ arrives at the depot. The problem consists of assigning each order to a drone flight $k \in \mathcal{K}$ and determining the optimal sequence of stops, such that:

- all orders are served,
- payload and energy constraints are respected (Section 2.3),
- total weekly operating cost C_{total} is minimized (Section 2.2).

The number of drones $M = |\mathcal{K}|$ is the sole decision variable. All routes are closed loops departing from and returning to the depot.

2.2 Objective Function

The model minimizes the total weekly operating cost, with the number of drones M as the sole decision variable (the number of bikers is fixed at $N_b = 1$).

$$\min_M C_{\text{total}} = C_{\text{capex}} + C_{\text{energy}} + C_{\text{labor}} + C_{\text{maintenance}} + C_{\text{facility}} \quad (1)$$

Capital expenditure (amortized weekly): Each drone costs $c_d = 13,000$ CHF and has a mechanical lifetime of 4,000 flight hours. Let H denote weekly operating hours:

$$C_{\text{capex}} = \frac{H \cdot c_d}{4,000} \cdot M \quad (2)$$

This yields 214 CHF/drone/week under Scenario 1 ($H = 100$ h) and 115 CHF/drone/week under Scenario 2 ($H = 42$ h).

Energy cost: Let D [km] be the total distance flown per week and Δh [m] the total absolute altitude change per week, both computed exogenously by the routing module. The weekly energy cost is:

$$C_{\text{energy}} = (52 \cdot D + 3.6 \cdot D \cdot \bar{w} + 0.12 \cdot |\Delta h|) \times 0.00027 \quad [\text{CHF}] \quad (3)$$

where $\bar{w} = 1$ kg is the assumed average payload weight and 0.00027 CHF/Wh is the Swiss electricity tariff. The derivation of the energy coefficients is detailed in Appendix A.

Labor cost: One supervisor monitors up to five drones simultaneously; one biker operates at all times:

$$C_{\text{labor}} = \frac{M}{5} \cdot H \cdot 45 + H \cdot 25 \quad [\text{CHF/week}] \quad (4)$$

Maintenance cost: Based on 3,000 CHF per drone per year, which include insurance and software license:

$$C_{\text{maintenance}} = \frac{3,000}{52} \cdot M \approx 57.7 \cdot M \quad [\text{CHF/week}] \quad (5)$$

Warehouse cost: Based on the previous assumptions and on the numbers in Appendix:

$$C_{\text{facility}} = 34.62 + 530.77 = 565.38 \text{ CHF/week} \quad (6)$$

The complete objective function is:

$$C_{\text{total}} = \frac{H \cdot c_d}{4,000} \cdot M + (52D + 3.6 D\bar{w} + 0.12 |\Delta h|) \cdot 27 \cdot 10^{-5} + \left(\frac{M}{5} \cdot H \cdot 45 + H \cdot 25 \right) + 57.7 \cdot M + 565.38 \quad (7)$$

2.3 System Constraints and Fleet Dimensioning

The following constraints govern every drone flight in the network.

1. **Closed-loop depot return.** Every drone departs from and returns to the drone facility after each delivery mission:

$$x_{0j} = x_{i0} = 1 \quad \forall i, j \in \mathcal{V} \quad (8)$$

This ensures battery packs can be swapped and drones reloaded for the next optimization slot.

2. **Payload capacity.** The drone carries a mandatory 3 kg cargo container, leaving 12 kg of usable payload per flight shared across all consolidated orders:

$$\sum_{j \in \mathcal{P}_k} w_j \leq 12 \quad [\text{kg per flight}] \quad (9)$$

where w_j is the weight of order j on flight k .

3. **Battery and energy autonomy.** Total energy consumed along a flight path must not exceed the usable battery capacity:

$$E_{\text{flight}} = \sum_{(i,j) \in A_k} (52 \cdot D_{ij} + 3.6 \cdot D_{ij} \cdot \text{load}_{ij} \cdot \bar{w} + 0.12 \cdot |\Delta h_{ij}|) \leq 3,000 \quad [\text{Wh}] \quad (10)$$

where D_{ij} [km] is horizontal distance, $|\Delta h_{ij}|$ [m] is absolute altitude change, and load_{ij} is the payload weight carried on segment (i, j) .

4. **Temporal availability.** A drone dispatched at time t becomes available again only after completing its full service cycle:

$$t_k^{\text{available}} = t_k^{\text{dispatch}} + T_k^{\text{flight}} + 2 \cdot |\mathcal{P}_k| + 5 \quad [\text{min}] \quad (11)$$

where T_k^{flight} is the total flight time of route k , $|\mathcal{P}_k|$ is the number of stops, 2 min is the drop-off time per stop, and 5 min is the post-flight battery swap and reload time at the facility.

5. **Fleet dimensioning.** The minimum fleet size M is set to absorb peak demand without missed deliveries:

$$M \geq 1.2 \cdot \frac{\text{Peak hourly demand [orders/h]}}{\text{Drone hourly capacity [orders/drone} \cdot \text{h]}} \quad (12)$$

3 Data description and manipulation

3.1 Topography

To compute the distances, we took several data:

- **Center of communes:** We assumed the GPS point given on Wikipedia for each commune as roughly the center in term of population.
- **Distances:** Using the haversine [19] library in Python, we can compute the distances in between all these centers, to have our base distances.
- **In-commune and randomized distances:** For deliveries in the same commune, the distance should not be zero as the commune are quite vast and the distances in between each localities are not negligible (e.g, the distance in between Niouc and Zinal in Anniviers is over 16km), we took the half of the longest line in each commune as the commune's radius. Then we randomized, for each 15-minutes round, the distances this way:

$$\text{Distance}_{A \rightarrow B} + \text{random distance}$$

where random distance $\in [\frac{r_B}{2}, r_B]$, with r_B the arrival commune's radius.

- **Altitudes:** Using the interactive maps of Canton Wallis [20], we could find the minimum and maximum altitudes of the communes. We focused on populated areas, so the peaks of mountains are not taken into account in the maxima. Then, for the simulations, we generate a random altitude using the minimum and maximum as boundaries.

3.2 Demand

The demand model combines Swiss food delivery market data, regional population metrics, and restaurant activity patterns to project the precise spatiotemporal distribution of orders across Sierre.

3.2.1 Baseline Weekly Demand per Person

Using data from the Just Eat Swiss market report [12], the baseline per-capita weekly order frequency is calculated by summing across all ordering-frequency categories k :

$$\text{Orders per person per week} = \sum_k \text{Share}_k \times \text{Frequency}_k \quad (13)$$

Substituting the empirical proportions yields a baseline frequency of ≈ 0.332 weekly orders per individual, accounting for the 20.1% non-ordering segment:

$$\begin{aligned} \text{Orders per person per week} &= 0.029(3) + 0.095(1) + 0.161 \left(\frac{2.5}{4.3} \right) \\ &\quad + 0.168 \left(\frac{1}{4.3} \right) + 0.145 \left(\frac{1}{4.3 \times 2.5} \right) \\ &\quad + 0.088 \left(\frac{1.5}{52.1} \right) + 0.113 \left(\frac{0.5}{52.1} \right) + 0.201(0) \\ &\approx 0.332 \end{aligned} \quad (14)$$

3.2.2 Commune-Level Fast-Food Demand

Permanent resident populations for the ten target communes were gathered from the Valais Statistical Office’s STATPOP database [11]. The baseline weekly demand for commune i (D_i^{week}) is computed as follows:

$$D_i^{\text{week}} = \text{Population}_i \times 0.332 \quad (15)$$

Because this network exclusively supports fast-food logistics (McDonald’s, Burger King, and Wok Royal), a 50% adjustment factor is applied to isolate this specific market share:

$$D_i^{\text{fast food}} = 0.5 \times D_i^{\text{week}} \quad (16)$$

3.2.3 Daily and Hourly Disaggregation

Daily demand ($D_{i,d}$) is allocated using temporal profiles extracted from existing household-cooked meal logistics literature [13], mapping daily share indices (Share_d) across the week:

$$D_{i,d} = D_i^{\text{fast food}} \times \text{Share}_d \quad (17)$$

Hourly activity distributions are captured by sampling relative visitor affluence levels from Google Maps for the target establishments. The mean hourly affluence (Affluence_h) is normalized to establish the active hourly shares (Share_h):

$$\text{Affluence}_h = \frac{\text{Affluence}_{\text{McDonald's},h} + \text{Affluence}_{\text{Burger King},h} + \text{Affluence}_{\text{Wok Royal},h}}{3} \quad (18)$$

$$\text{Share}_h = \frac{\text{Affluence}_h}{\sum_{h=11}^{22} \text{Affluence}_h} \quad (19)$$

Combining these spatial and temporal parameters yields the deterministic hourly order volume per commune ($D_{i,d,h}$):

$$D_{i,d,h} = D_{i,d} \times \text{Share}_h \quad (20)$$

3.2.4 Stochastic Disaggregation

To mirror realistic, independent delivery requests within any given hour, the deterministic value $D_{i,d,h}$ is disaggregated into four discrete 15-minute intervals following a stochastic Poisson arrival process. This final resolution step ensures that the temporal fluctuations match the exact courier pickup schedule of one drone route cycle every 15 minutes.

4 Results

The baseline scenario represents the reference configuration of the drone delivery system. It combines the demand estimation, the centralized hub structure, the bike-based restaurant collection process, and the drone delivery routes. The objective of this section is therefore to interpret what the model results imply for the proposed business model in the Sierre District. Table 1 summarizes the main results obtained for the baseline scenario.

For clarity in the subsequent economic formulations, a structural distinction is made between a *delivery* and a *flight*: **delivery** refers strictly to an individual customer order payload, whereas a **flight** denotes a single consolidated aircraft mission, which may aggregate up to 10 distinct customer deliveries per sortie.

Key indicator	Baseline scenario
Meals delivered	8,376
Missed deliveries	0
Operating drones	30
Number of weekly flights	952
Average deliveries per flight	8.80
Total distance flown	29,876.31 km
Average distance per delivery	3.57 km
Total energy consumed	1.96 MWh
Weekly operating cost	42,094.39 CHF
Average cost per delivery	5.02 CHF

Table 1: Main results for the baseline scenario

4.1 Service performance

The first result concerns the ability of the system to serve the estimated demand. In the baseline scenario, 8,376 meals are delivered over one week, with no missed deliveries. This indicates that, under the assumptions of the model, the fleet size and routing structure are sufficient to absorb the expected weekly demand.

From a service-quality perspective, this is a strong result. Food delivery is highly sensitive to reliability, and missed orders would directly reduce customer satisfaction. The absence of missed deliveries therefore suggests that the proposed system can provide a reliable service level for the selected communes.

4.2 Route consolidation and fleet utilization

The second result concerns route consolidation and fleet utilization. With a fleet of 30 operating drones, the system performs 952 flights over the week to deliver 8,376 meals. This corresponds to an average of 8.80 deliveries per flight.

This shows that the model does not rely on individual point-to-point drone trips, instead, orders are bundled into consolidated drone flights. This is consistent with the routing and capacity logic seen in class: the objective is not only to reach customers, but also to assign demand efficiently to vehicles while respecting operational constraints. In this case, consolidation is essential because it spreads flight costs over several deliveries and improves the use of the deployed drone fleet.

The average of 8.80 deliveries per flight is close to the operational consolidation assumption of around 10 client orders per flight. This suggests that the system achieves a high level of route consolidation, even though not every flight reaches the capped loading assumption. This is expected in practice, since orders are spatially dispersed across communes and arrive at different moments during the day. Therefore, the effective loading of each flight is constrained not only by the drone’s payload capacity, but also by routing feasibility, demand timing, range, and operational constraints.

4.3 Distance and energy performance

The total weekly distance flown is 29,876.31 km. While this number appears high, it should be interpreted together with the average distance per delivery, which is around 3.57 km. This indicates that the central hub location allows the system to serve a large number of orders with relatively short average delivery distances.

From a logistics perspective, this supports the relevance of using a drone hub in Sierre to connect restaurants in the valley with surrounding communes. The hub structure limits the average distance per delivery while still allowing the system to reach areas where conventional road-based delivery is less efficient.

The total energy consumption reaches 1.96 MWh over the week. This value remains limited relative to the number of deliveries performed, mainly because orders are consolidated into multi-delivery flights rather than served through individual trips. From a sustainable

logistics perspective, this reinforces the importance of route consolidation and short average delivery distances in reducing energy use per delivered meal.

4.4 Economic performance

The weekly operating cost reaches 42,094 CHF, corresponding to an average cost of 5.02 CHF per delivery. Since the objective function minimizes total weekly operating cost, this result is central to evaluating the economic performance of the proposed business model. The total cost reflects the combined effect of drone amortization, energy consumption, labor, maintenance, and facility-related expenses. The average cost per delivery is therefore not optimized directly, but obtained by spreading the weekly operating cost over the 8,376 meals delivered.

This result is particularly relevant when compared with current delivery prices in the region. As shown in Appendix A.4, an Uber Eats delivery from McDonald’s in Sierre to Zinal charges a delivery fee of 17.90 CHF. The average delivery cost obtained in the baseline scenario, 5.02 CHF, is therefore substantially lower than this observed delivery fee. This comparison should be interpreted carefully, since the Uber Eats price includes platform pricing decisions and not only operational costs. Nevertheless, it suggests that a consolidated drone-based delivery model could offer a competitive cost structure for long-distance food delivery in mountainous areas.

This unit cost is one of the most important economic indicators of the model. It suggests that the system could be competitive if demand is sufficiently high and if the deployed drone fleet is sufficiently utilized. However, the result should be interpreted carefully: the weekly cost is not only driven by the number of deliveries. A large part of the cost comes from resources that must be available for the system to operate, such as the hub, drone maintenance, drone amortization, and supervision capacity. These costs are spread over all delivered meals. As a result, the average cost per delivery improves when the system is used intensively, and worsens when demand or fleet utilization is too low.

4.5 Key Takeaways

The baseline scenario shows that the proposed drone delivery system can operate effectively. It delivers all estimated orders, generates no missed deliveries, and reaches a competitive average cost.

The main conclusion is that the business model appears promising when demand is high enough to keep the drone fleet well utilized and when orders can be consolidated into multi-delivery flights. However, this performance depends on key operational conditions: operating hours, demand level, and fleet size.

The following sensitivity analysis therefore tests whether the system remains efficient when these conditions change.

5 Sensitivity Analysis

The sensitivity analysis evaluates the robustness of the baseline configuration across two alternative scenarios, varying operating hours and demand level.

1. Table 2 compares the baseline (all restaurant’s operating [100 hours], full demand) against Scenario 1 (peak hours only [42 hours], full demand).
2. Table 3 compares Scenario 2 (100 hours, half demand, half fleet) against Scenario 3 (peak hours only, half demand, half fleet).

5.1 Operating hours: Baseline vs. Scenario 1

Table 2: Full demand: baseline vs. peak hours only

	Baseline (100h)	Scenario 1 (peak only)
Meals delivered	8,376	4,840
Missed deliveries	0	0
Flights	952	527
Avg deliveries per flight	8.80	9.18
Total energy consumed	1.96 MWh	1.16 MWh
Weekly operating cost	42,094 CHF	19,113 CHF
Avg cost per delivery	5.02 CHF	3.95 CHF
Total distance flown	29,876 km	17,402 km
Avg distance per delivery	3.57 km	3.60 km
Avg distance per flight	31.38 km	33.02 km

Main results: Restricting operations to peak hours only reduces the weekly cost by 55%, from 42,094 CHF to 19,113 CHF, while serving 42% fewer meals. The average cost per delivery drops from 5.02 CHF to 3.95 CHF, meaning the system becomes more cost-efficient per order despite handling less volume. This is explained by higher flight consolidation under peak-hour operation: average deliveries per flight rise from 8.80 to 9.18, as demand is concentrated in a narrower time window. Average distances per flight and per delivery remain nearly identical across both scenarios (3.57 km vs. 3.60 km), confirming that the spatial structure of demand is stable regardless of operating hours. The key trade-off is coverage: Scenario 1 leaves off-peak demand unserved, which may be acceptable for a cost-focused entry strategy but limits market reach.

5.2 Demand sensitivity: Scenarios 2 and 3

Table 3: Half demand, half fleet: 100h vs. peak hours only

	Scenario 2 (100h)	Scenario 3 (peak only)
Meals delivered	4,088	2,376
Missed deliveries	0	0
Flights	498	285
Avg deliveries per flight	8.21	9.21
Total energy consumed	789 kWh	493 kWh
Weekly operating cost	22,539 CHF	10,351 CHF
Avg cost per delivery	5.51 CHF	4.35 CHF
Total distance flown	12,614 km	7,810 km
Avg distance per delivery	3.09 km	3.28 km
Avg distance per flight	25.33 km	27.40 km

Main results: Halving both demand and fleet size produces results that scale almost linearly with the baseline. Under Scenario 2 (100 hours, half demand), the weekly cost falls to 22,539 CHF, roughly half of the baseline, and the average cost per delivery rises slightly to 5.51 CHF. This small increase reflects the fixed cost components of the model (supervision, maintenance, facility) that do not scale proportionally with demand. Under Scenario 3 (peak hours, half demand), the weekly cost drops to 10,351 CHF with an average cost of 4.35 CHF per delivery, the lowest across all scenarios. Notably, zero missed deliveries are recorded in every scenario, confirming that the system remains operationally feasible even under reduced capacity. The average distance per delivery decreases under half-demand conditions (3.09–3.28 km vs. 3.57–3.60 km), suggesting that at lower demand the routing module selects shorter, more local routes as distant communes generate fewer orders.

5.3 Key Takeaways

Across all four configurations, the system never misses a delivery and maintains average costs between 3.95 and 5.51 CHF per order. The results demonstrate that the model is robust to both demand reduction and operating hour constraints. The most cost-efficient configuration is peak-hour operation (Scenario 1 and 3), driven by higher consolidation and lower fixed cost exposure. The sensitivity to demand is moderate: halving demand raises the unit cost by less than 10% in the full-hours case, which suggests the fixed cost structure is well-absorbed at the demand levels considered.

6 Conclusion

6.1 Limitations

Biker’s operational bottleneck: We modelled the process of bringing the packaged food to the depot, swapping batteries and loading the package in the drones as a fixed-time one-person task, whilst in practice this could take different duration and

could need more than one operator. However we set the parameter to a realistic conservative value.

Zone specific results: The district of Sierre has very particular topographic characteristics. A moderate number of communes at close line-air distance but with significantly difficult reachability. Our model might have lower performances in other district where the distance between communes is larger.

Robustness of assumptions: Our belief of high desirability of a delivery system operated by drones might be actually proven wrong: Valais is one of the most conservative Cantons in Switzerland and while the idea of a better service for a lower price should incentivize acceptance, the idea of a piloted drone reaching your house can raise doubts. Furthermore, we assume that the development of new regulations will successfully drive technical innovation in this sector, but this stands on uncertain grounds.

Exclusion of Financing Costs and Macroeconomics: Our cost function assumes the project is fully funded upfront. To keep the model straightforward, we did not take into account the cost of capital, bank interest rates, inflation, or Swiss corporate tax variations over the lifecycle. In a real-world corporate setup, these financing costs and macroeconomic factors would alter the net present value (NPV) and the total cost of ownership.

6.2 Key Takeaways

This project assessed whether a drone-based last-mile delivery system could be operationally and economically viable in the mountainous district of Sierre. The answer is yes, and the reasons are specific to this context.

The problem is structural, not technical. The communes surrounding Sierre are not far in straight-line distance, the average mission is around 30 km. They are underserved because road complexity makes conventional delivery economically unviable. Drones do not need to be better than cars in general. They only need to be better than cars in this specific terrain, and they are.

The economics are compelling. The baseline scenario delivers 8,376 meals per week with zero missed deliveries at an average cost of 5.02 CHF per delivery, which stands in stark contrast to the 17.90 CHF retail delivery fee currently observed on Uber Eats for the same geography. This spread provides the enterprise with an exceptional pricing runway and strategic maneuver room.

Consolidation is the mechanism. The cost efficiency of the system does not come from drones being cheap to operate. It comes from bundling 8 to 9 orders per flight, which spreads fixed costs across multiple deliveries. This is the central insight of the routing model: the drone hub in Sierre acts as a consolidation point, and its value grows with demand density.

The entry strategy is clear. Restricting operations to peak hours only reduces weekly cost by 55% while improving the average cost per delivery from 5.02 CHF to 3.95 CHF.

A real operator should launch during peak hours, and expand operating hours only as demand justifies it.

The system is robust. Across all four scenarios tested, the system never misses a delivery. Halving demand and fleet size raises the unit cost by less than 10%, confirming that the fixed cost structure is well-absorbed even under adverse conditions. The main risks to this model are not operational, they are regulatory acceptance and social adoption in a context where both remain uncertain.

6.3 Suggestions for Improvement

Dynamic fleet sizing. The model optimizes a fixed fleet size over a weekly horizon. A natural extension would be to solve the fleet sizing problem at a finer time granularity (hourly or by 15-minute slot), allowing the number of active drones to vary with demand. This would reduce idle asset time during off-peak hours and lower the average fixed cost per delivery. A drone rental model, rather than outright purchase, could make this dynamic sizing practically achievable by removing the capital commitment of a fixed fleet and allowing operators to scale capacity up or down in response to realized demand.

Multi-hub architecture. The current system relies on a single centralized hub in Sierre. For communes at the edge of the service area such as Anniviers, a secondary relay hub could reduce average flight distance, lower energy consumption per delivery, and extend the serviceable range of the network without increasing fleet size.

Biker bottleneck modelling. The current model fixes the biker’s collection and loading process as a single-person task with a constant 15-minute cycle. In practice, during peak hours the biker may become the binding constraint on fleet throughput rather than the drones themselves. A more granular model of the collection and battery-swapping process, potentially with a second operator during peak slots, could unlock additional delivery capacity without increasing fleet size.

Weather and seasonality. The model assumes constant operating conditions. In the Sierre-Anniviers district, Alpine wind patterns and winter conditions significantly affect drone performance and regulatory flight windows. Incorporating seasonal demand and weather-adjusted energy parameters (battery capacity drops 15–20% at -10°C) would improve the robustness of the cost estimates across the full yearly operating cycle.

Business model development: By driving the unit economics down to approximately 4 CHF per delivery, significantly below the current market pricing benchmarks of over 15 CHF, the initial operational framework demonstrates a highly competitive cost advantage. However, capturing this margin requires a comprehensive evaluation of the broader business model development. Future iterations of this work must formalize the overarching financing structures, multi-stage scaling vectors, and dynamic revenue architectures needed to successfully transition this pilot network into a commercially viable, long-term enterprise.

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A Appendix

A.1 Energy Model

The per-segment energy consumption of the drone is modeled as:

$$E_{ij} = \alpha \cdot D_{ij} + \beta \cdot D_{ij} \cdot \text{load} \cdot w + \gamma \cdot |\Delta H_{ij}| \quad (21)$$

where D_{ij} [km] is the horizontal distance, $\text{load} \in \{0, 1\}$ is a binary indicator equal to 1 when the drone carries cargo, w [kg] is the payload weight, and $|\Delta h_{ij}|$ [m] is the absolute altitude difference between i and j . Both ascent and descent are assumed to consume energy at the same rate, as electric braking dissipates energy as heat rather than recovering it. The three parameters α , β , γ are derived from the T-Drones M1500 manufacturer specifications as follows.

Datasheet inputs

- Battery: 4× ARES 6S 30 Ah Li-ion; nominal voltage = $6 \times 3.7 = 22.2$ V
- Drone mass with battery: 19.9 kg; maximum payload: 15 kg; MTOW: 34.9 kg
- Hover endurance: ≥ 60 min at 5 kg payload; ≥ 30 min at 15 kg payload
- Maximum horizontal speed: 13 m/s; cruise assumed at 30 km/h (65% of max)
- Maximum ascent speed: 5 m/s

Step 1 - Usable battery energy

$$E_{\text{gross}} = 4000 \text{ Wh} \quad (22)$$

Applying 75% usable depth of discharge (standard practice for Li-ion longevity):

$$E_{\text{usable}} = 4000 \times 0.75 = 3000 \text{ Wh} \quad (23)$$

Step 2 - Deriving α (empty-drone energy per km)

At 5 kg payload the drone hovers 60 min on E_{usable} , giving an average power draw of 3000 W. Power scales with total lifted mass via momentum theory (exponent 1.5):

$$\frac{P_0}{P_5} = \left(\frac{19.9}{24.9} \right)^{1.5} \implies P_0 \approx 1,572 \text{ W} \quad (24)$$

Endurance at empty: $3000 / 1572 = 1.91$ /h. Range at cruise speed:

$$R_0 = 1.91 \text{ h} \times 30 \text{ km/h} = 40.7 \text{ km} \quad (25)$$

$$\alpha = \frac{E_{\text{usable}}}{R_0} = \frac{3000}{40.7} \approx \mathbf{52 \text{ Wh/km}} \quad (26)$$

Step 3 - Deriving β (payload penalty per km per kg)

Applying momentum theory to both payload scenarios:

Scenario	Total mass (kg)	Power (W)	Energy/km (Wh/km)
5 kg payload	24.9	2,131	71.0
15 kg payload	34.9	3,214	107.1

$$\beta = \frac{107.1 - 71.0}{15 - 5} \approx \mathbf{3.6 \text{ Wh/km/kg}} \quad (27)$$

Step 4 - Deriving γ (altitude energy per metre)

Climbing is work against gravity. At maximum ascent speed with full MTOW:

$$P_{\text{climb}} = m \cdot g \cdot v = 34.9 \times 9.81 \times 5 = 1,712 \text{ W} \quad (28)$$

Energy per metre of altitude change:

$$\frac{E}{|\Delta H|} = \frac{1,712 \text{ W}}{5 \text{ m/s} \times 3,600 \text{ s/h}} = 0.095 \text{ Wh/m} \quad (29)$$

Accounting for motor efficiency ($\eta \approx 85\%$):

$$\gamma \approx \mathbf{0.12 \text{ Wh/m}} \quad (30)$$

Summary

Parameter	Value	Unit	Derivation basis
α	52	Wh/km	Empty-drone hover endurance
β	3.6	Wh/km/kg	5 kg vs. 15 kg hover comparison
γ	0.12	Wh/m	Physics: $mg \Delta H $ / motor efficiency

A.2 Facility Operational Scenarios

To integrate the physical infrastructure requirements of the network, two distinct facility deployment strategies are evaluated under the system assumptions:

- **Renting a Strategic Hub**

This option involves leasing an existing industrial garage or small commercial warehouse space (approximately 50–100 m²) within the artisanal or commercial zones of Sierre, specifically optimized for its immediate physical proximity to the primary restaurant partners [5].

- **CAPEX:** A one-time initial investment of 18,000 CHF is allocated for high-voltage industrial grid outfitting, automated safety ventilation, and security access retrofitting inside the rented space, amortized over 10 years. [6].
- **OPEX:** Ongoing fixed costs are budgeted at 27,600 CHF per year (calculated as 1,800 CHF/month for baseline commercial rent plus 500 CHF/month for commercial utility drawings and general facility maintenance contracts).

A.3 Technical and Economic Specifications Matrix

The following table compiles the complete set of hardware, operational, and financial specifications extrapolated from our core drone proxy platform and local economic indices.

Table 4: Industrial UAV Fleet Operational and Financial Baseline Specs

Parameter Category	Specification Metric	Model Baseline Value
<i>Physical & Hardware</i>	Aircraft Platform Proxy	T-Drones M1500 Hexacopter
	Maximum Payload Capacity	15 kg
	Consolidated Target Load (80% Utilization)	12 kg
	Average Customer Order Weight ($\bar{w}$)	1 kg
	Safe Landing Clearance Radius	1.5 meters
	Enclosure Protection Rating	IP55 (Weatherproof)
<i>Energy & Battery</i>	Dual-Pack Battery Total Capacity	~ 3.70 to 4.00 kWh
	Industrial Charging Power Tariff	0.27 CHF/kWh
	Battery Swapping Methodology	Mechanical Hot-Swap
<i>Flight Constraints</i>	Mixed Operational Range (Mountain Terrain)	35 km to 55 km
	Certified Airframe Lifespan	4,000 Flight Hours
	Expected Structural Lifetime (Scenario 1)	~ 1 Year
	Expected Structural Lifetime (Scenario 2)	~ 2 Years
<i>Financial CAPEX</i>	Base Airframe Capital Investment	11,000 CHF / unit
	Smart Battery Pack Procurement	2,000 CHF / pack
	Initial Depot High-Voltage Setup	18,000 CHF
<i>Personnel & OPEX</i>	Scheduled Annual Maintenance (MRO)	3,000 CHF / drone / year
	Remote Fleet Supervision Ratio ($M : N$)	1 Supervisor : 5 Drones
	Central Fleet Operator Wage	45 CHF / hour
	Fixed Bicycle Support Courier Wage ($N_b = 1$)	25 CHF / hour

A.4 Uber Eats delivery price benchmark

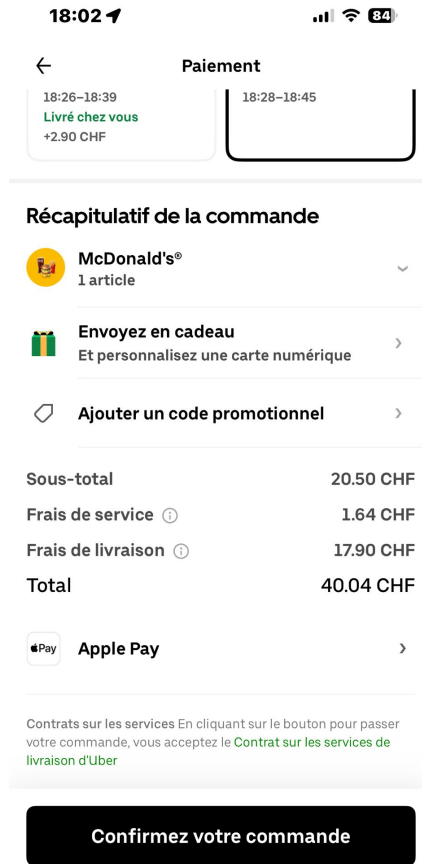


Figure 2: Uber Eats delivery fee from McDonald's Sierre to Zinal, showing a delivery fee of 17.90 CHF.

A.5 Pixel counting trick

To compute the hourly demand factor for each restaurant each day, we compare each affluence with the maximal one. Then, using Darboux integrals, we took the ratios

$$\frac{\text{area under the curve hour}}{\text{AUC daily curve}}$$

to have the daily percentage of the demand, for each restaurants.

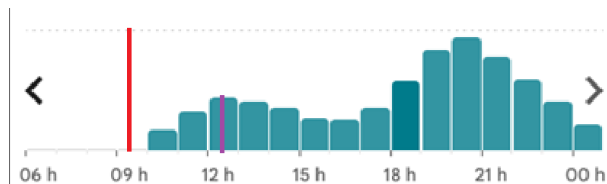


Figure 3: The highest weekly demand and Friday affluence at 12 for Sierre's McDonald's